



Hydraulic & Engineering Instruments

Progressive in Design Perfect in Performance

OPERATING MANUAL



An ISO 9001 : 2015 Company



OPERATING INSTRUCTIONS

FOR

“L Box Test on Self Compacting Concrete”

Hydraulic & Engineering Instruments
(An ISO 9001:2015 Company)

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EQUIPMENT

The equipment comprises the following: -

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L Box Test on Self Compacting Concrete

This test for self compacting concrete is based on a design for underwater concrete. The test assesses the flow of the concrete and also the extent to which it is subjected to blocking by reinforcement. The apparatus is shown in the figure.

The apparatus consist of rectangular section box in the shape of an 'L', with a vertical and horizontal section, separated by a movable gate, in front of which vertical length of reinforcement bar are fitted. The vertical section is filled with concrete, and then the gate lifted to let the concrete flow into the horizontal section.

When the flow has stopped, the height of the concrete at the end of the horizontal section is expressed as a proportion of that remaining in the vertical section. It indicates the slope of the concrete when at rest. This is an indication passing ability, or the degree to which the passage of concrete through the bars is restricted.

The horizontal section of the box can be marked at 200mm and 400mm from the gate and the times taken to reach these points measured. These are known as the T20 and T40 times and are an indication for the filling ability.

Assessment of test:

This is a widely used test, suitable for laboratory and perhaps site use. It asses filling and passing ability of SCC, and serious lack of stability (segregation) can be detected visually. Segregation may also be detected by subsequently sawing and inspecting sections of the concrete in the horizontal section.

Equipment for L Box Test

- L box of a stiff non absorbing material

- Trowel

- Scoop

- Stopwatch

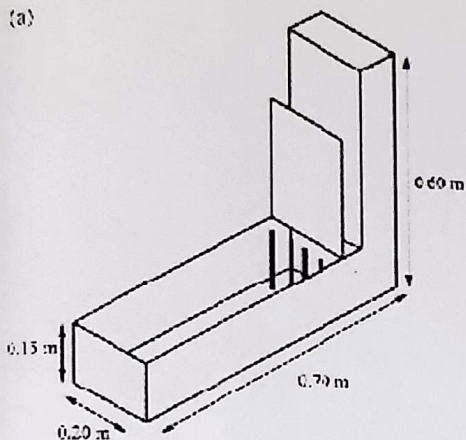


Fig.:L Box test Apparatus

Procedure of L Box Test

About 14 liter of concrete needed to perform the test, sampled normally. Set the apparatus level on firm ground, ensure that the sliding gate can open freely and then close it.

Moisten the inside surface of the apparatus, remove any surplus water, fill the vertical section of the apparatus with the concrete sample.

Leave it stand for 1 minute. Lift the sliding gate and allow the concrete to flow out into the horizontal section. Simultaneously, start the stopwatch and record the time for the concrete to reach the concrete 200 and 400 marks.

When the concrete stops flowing, the distances 'H1' and 'H2' are measured. Calculate $H2/H1$, the blocking ratio. The whole has to be performed within 5 minutes.

Interpretation of the result:

If the concrete flows as freely as water, at rest it will be horizontal, so $H_2/H_1=1$. Therefore the nearest this test value, the 'blocking ratio', is unity, the better the flow of concrete.